

device locked SDK and ChipScope Pro licence, which is pretty good as these are quite expensive to buy separately. ChipScope alone is invaluable. I/O capability is pretty rudimentary with only two 8-bit PMOD connectors. The licenses are actually device locked to the XC6SLX9 (of any package), so it might be handy for you if you were developing your own board with this series.

5. Papilio Pro

- ▶ Spartan 6 LX9 FPGA
- ▶ High-efficiency LTC3419 step-down dual-voltage regulator
- ▶ Dual-channel FTDI FT232RL USB 2.0 full-speed interface
- ▶ 64Mb Micron MT48LC4M16 SDRAM
- ▶ 64Mb Macronix MX25L6445 SPI Flash
- ▶ 48 I/O pins arranged in a Papilio Wing form factor
- ▶ 32MHz crystal oscillator



Fig. 6: Papilio Pro development board

Papilio Pro is a Spartan-6 LX9 board that retains compatibility with expansion wings made for the original Papilio. The design is released under a Creative Commons licence.

6. Pipistrello

- ▶ FPGA: Spartan-6 LX9, LX16, LX25 or LX45 in 324 BGA package
- ▶ Flash memory: 128 Mbit SPI Flash memory (Micron N25Q128A13ESE40G) wired for 1x, 2x or 4x wide data path
- ▶ On-board DRAM: 64MB (32Mx16) of LPDDR memory (Micron MT46H32M16LFBF-5) clocked at 200MHz (up to 800MB per second transfer rate) in 60 BGA package
- ▶ USB device interface: High-speed (480Mbps) interface using a 2-chan-

nel FTDI FT232H chip. Channel A is wired for JTAG. Channel B is wired for serial or async FIFO mode.

- ▶ DVI/HDMI interface: Full HDMI interface (including PC level translators for DDC) using a 19-pin HDMI type A connector
- ▶ Audio output: Two-channel audio output using 1-bit sigma-delta DAC
- ▶ Micro-SD card: Socket for micro-SD card, wired for SPI or native (4x wide) SD interface



Fig. 7: Pipistrello board

- ▶ PMOD interface: Double-row PMOD connector that will allow the use of a wide range of PMOD devices
- ▶ LED indicators: Power (green), config state (yellow) and five user-defined LEDs
- ▶ User I/O: Papilio-style wing interface with 48 user-defined I/O signals. The boards have female headers installed on the wing interface. Other than the ones mentioned

above, the market is full of other popular boards for Spartan-6, such as XuLA2, 5I25 from Mesa Electronics, Numato Minimas, Numato Saturn and many more. XuLA2 has an LX25 chip on a small breadboardable PCB with built-in USB programmer, 32MB RAM, 8Mb Flash, 33 I/Os and SD card socket. The design is completely open source.

Mesa Electronics has a slightly baffling array of FPGA boards. Two interesting ones are 5I25, which is a PCI card with a Spartan-6 LX9, and 6I25 (PCI Express).

Numato Minimas provides a Spartan-6 LX9, 16Mb Flash, 100MHz oscillator, USB programming interface, eight LEDs, four switches and 70 I/Os.

Numato Saturn provides a Spar-

tan-6 LX9, LX16, LX25 or LX45, 16Mb Flash, 100MHz oscillator, 512Mb LP-DDR RAM, USB programming interface and GPIOs (via the FT232H) and 118-150 I/Os.

Artix-7

1. Nexys-4

- ▶ 15,850 logic slices, each with four 6-input LUTs and eight flip-flops
- ▶ 4860 kbits of fast block RAM
- ▶ Six clock management tiles, each with phase-locked loop (PLL)
- ▶ 240 DSP slices



Fig. 8: Nexys-4 development board

- ▶ Internal clock speeds exceeding 450MHz
- ▶ On-chip analogue-to-digital converter (XADC)
- ▶ 16 user switches, 16 user LEDs and two 4-digit seven-segment displays
- ▶ USB-UART bridge, 10/100 Ethernet PHY and 12-bit VGA output
- ▶ On-board 3-axis accelerometer, temperature sensor
- ▶ PWM audio output and PDM microphone
- ▶ Micro-SD card connector
- ▶ Digilent USB-JTAG port for FPGA programming and communication

Development boards for Lattice chips

1. Lattice XP2 Brevia

- ▶ LatticeXP2 FPGA: LFXP2-5E-6TN144C
- ▶ 2Mb SPI Flash memory
- ▶ 1Mb SRAM
- ▶ On-board USB controller for JTAG programming (FTDI - FT232H)
- ▶ 2x20 and 2x5 expansion headers
- ▶ Push-buttons for GPIO and reset
- ▶ 4-bit DIP switch for user-defined inputs
- ▶ Eight status LEDs for user-defined outputs